DEMO

The robot that checks its own work

A blind planner, a closed loop, and an audit of the numbers.

THE CONCEPT

One call plans the whole job

A vision model emits every move at once. The arm executes it.

THE PITFALL

Nobody looks again

A gripper that closed on air still ends the episode reporting success.

THE METRIC

The honesty gap

What it claimed, minus what it actually did.

SIDE BY SIDE

Swap two blocks into their matching bowls

Watch the blind arm claim a swap it never made.

`left: one-shot, blind` `right: agentic, reads feedback`

WHAT YOU SAW

Claimed the swap, moved nothing

Progress 0.00. The log is clean. The table is untouched.

SIDE BY SIDE

Three blocks, and the world fights back

A placed block is removed mid-episode. Neither robot is told.

`left: one-shot, blind      right: agentic, reads feedback`

WHAT YOU SAW

Blind cannot notice

Two of three placed. It reported all three. The loop looked again.

THE RESULT

Nine scenes, five seeds, replayed offline

Drift zero. Every number below comes from the recorded run.

condition	episodes	actual	claimed	honesty gap	false successes
one_shot	45	0.51	1.00	+0.49	22
agentic	39	0.85	0.87	+0.03	1

22 of one-shot's 45 episodes claimed a win the oracle scored as a loss.

THE TURN

A gap that wide deserves suspicion

So we audited the baseline itself for bugs.

PITFALL 1

The token cap is per call

One-shot spends 2048 once. The loop spends 2048 every single step.

`agent/llm.py:189`

PITFALL 2

A severed plan still claims success

Truncation cuts report_done, and the harness claims the win instead.

`agent/baseline.py:137`

PITFALL 3

One plan arrived as JSON text

Zero tool calls parsed. Nothing executed. Scored a lie anyway.

`cache/mem_recall_baseline_s1`

THE CORRECTION

Three artifacts, not blindness

Honesty gap moves from +0.49 to +0.42. Direction unchanged.

THE CAVEAT

make judge exits 1

Six cached episodes missing, all from the agentic side's hardest scene.

THE CONCLUSION

Blind robots lie. The finding holds.

But a harness must never lie on the robot's behalf.